Lab exercises: Selection

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Tools for this lab: relational operators, `&&`, `||`, `!`, `bool`, and single `if` statements. Keep all code in `main()`. Use separate `if` statements where needed.

Exercise 1: Evaluate the conditions

Focus: relational operators, logical operators, and precedence.

```
int x = 8, y = 3, z = 8;
bool ready = false;
```

Complete the table without running a program. Write both the Boolean result and its numeric value (0 or 1).

	Expression	true / false	0 / 1
a	`x > y`	TRUE	1
b	`x != z`	FALSE	0
c	`(y >= 1) && (y <= 5)`	TRUE	1
d	`(x < y) || (z == 8)`	TRUE	1
e	`!ready`	TRUE	1
f	`(x < y) || (z == 8) && (y > 5)`	FALSE	0

(g) Explain which logical operator is evaluated first in (f).

The logical AND (&&) operator is evaluated first.

Exercise 2: Trace independent if statements

Focus: execution order and statements outside the if body.

Assume the following code is inside `main()`.

```
int  score;
int bonus = 0;
cin >> score;
if (score >= 50)
{
    bonus += 5;
    cout << "Pass\n";
}
if (score >= 80)
{
    bonus += 3;
    cout << "Excellent\n";
}
cout << "Bonus = " << bonus << "\n";
```

(a) Write the exact output for each input: **49, 50, 79, and 80**.

Input 49

Output : Bonus = 0

Input 50

Output : Bonus = 5

Input 79

Output : Bonus = 5

Input 80

Output : Bonus = 8

(b) For an input of 80, how many `if` bodies execute? Explain why.

Both the if bodies execute. Because 80 is greater than 50 and equal or more than 80.

(c) Does the final `cout` statement execute when the input is 49? Explain.

Yes it executes. The cout statement is outside the if blocks, so it always runs regardless of the input.

Exercise 3: Find and fix the logic errors

Focus: semicolons, equality, braces, and numeric ranges.

Each snippet is inside `main()`. For each part, explain the error and rewrite the affected statements so they match the intended behaviour.

(a) Low-stock warning. Display `Low stock` only when stock is less than 5. Always display `Check complete`. Test with 4 and 5.

```
int stock;
cin >> stock;
if (stock < 5);
    cout << "Low stock\n";
cout << "Check complete\n";
```

Error : There is no semicolon in if statement

Answer : if (stock < 5)

(b) PIN check. Only when the PIN equals 2468, display `Access granted` and set attempts to 0. For any other PIN, attempts must remain 3. Test both cases.

```
int pin, attempts = 3;
cin>>pin;
if (pin = 2468)
    cout << "Access granted\n";
    attempts=0;
cout << "Attempts left = " << attempts << "\n";
```

Error : Missing curly bracket, and use operator (=) instead of an equality comparison operator (==)

Answer :

```
if pin ( ==2468)
{
        cout << “Access granted \n”;
        attempts = 0;
}
```

(c) Valid mark. Display `Valid mark` only for marks from 0 to 100, inclusive. Test with -1, 0, 100, and 101.

```
int mark;
cin >> mark;
if (0 <= mark <= 100)
    cout << "Valid mark\n";
```

Error : C++ doesn't support chained ranges like (0 <= mark <= 100)

Answer :

```
int mark;
cin >> mark;
if (0 >= mark <= 100)
        cout << “Valid mark\n”;
```

Exercise 4: Campus shop discount

Focus: initialise a value, then update it using one if statement.

A campus shop gives a **10% discount** when a purchase is **RM100 or more**. For a smaller purchase, the discount is RM0.00. Assume the purchase amount is a valid non-negative number.

(a) Prepare pseudocode or a flowchart for the solution.



(b) Write a C++ program that reads the purchase amount, initialises the discount to zero, and uses **one if statement** to calculate any discount.

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

int main() {
    double purchaseAmount, discount = 0.00, amountPayable;

    cin >> purchaseAmount;

    if (purchaseAmount >= 100.00) {
        discount = purchaseAmount * 0.10;
    }

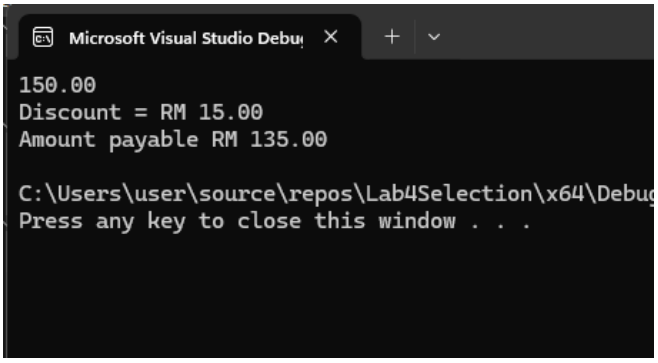
    amountPayable = purchaseAmount - discount;

    cout << fixed << setprecision(2);
    cout << "Discount = RM " << discount << "\n";
    cout << "Amount payable RM " << amountPayable << "\n";

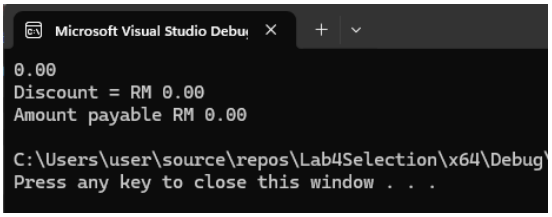
    return 0;
}
```

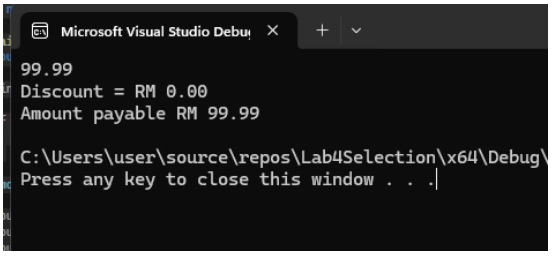
(c) Calculate and display the discount and final amount payable to two decimal places. Use `<iomanip>` for money formatting.

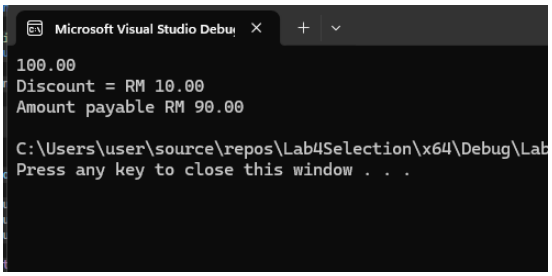
Sample input	Sample output (after the input prompt)
150.00	Discount = RM 15.00 Amount payable = RM 135.00

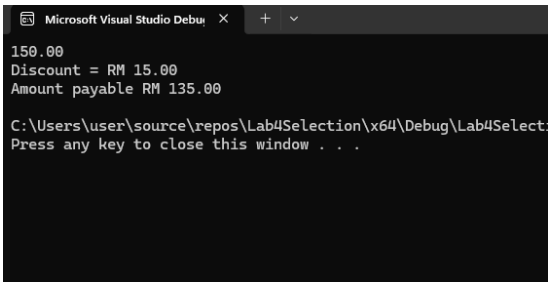


(d) Test your program with **RM0.00, RM99.99, RM100.00, and RM150.00**. Record the expected and actual results.







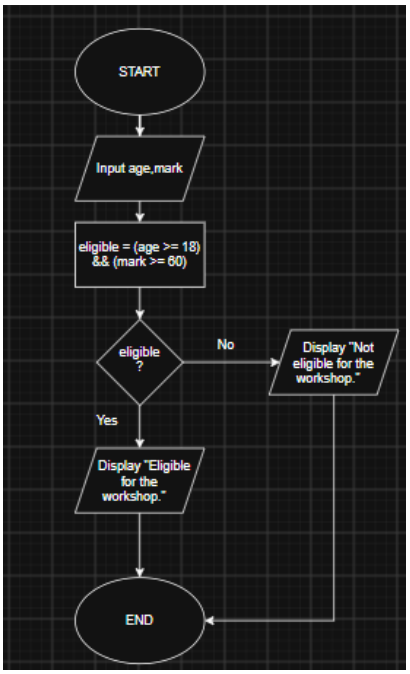


Exercise 5: Workshop eligibility checker

Focus: combine conditions with &&, store a bool, and use !.

A student can join a C++ workshop when **both** conditions are met: the student is **at least 18 years old** and has a prerequisite test mark of **at least 60**. Assume the age is non-negative and the mark is an integer from 0 to 100.

(a) Prepare pseudocode or a flowchart for the solution.



(b) Read the student's age and test mark. Declare a `bool` variable named `eligible` and assign it the result of a logical expression.

(c) Use `if (eligible)` to display **both** lines below in one body. Use braces.

```
Eligible for the workshop.
Please complete the registration form.
```

(d) Use a separate `if (!eligible)` statement to display `Not eligible for the workshop.`

```cpp
#include <iostream>
using namespace std;

int main()
{
    int age, mark;
    bool eligible;

    cout << "Enter age: ";
    cin >> age;

    cout << "Enter prerequisite test mark: ";
    cin >> mark;

    eligible = (age >= 18) && (mark >= 60);

    if (eligible)
    {
        cout << "Eligible for the workshop." << endl;
        cout << "You meet both requirements." << endl;
    }

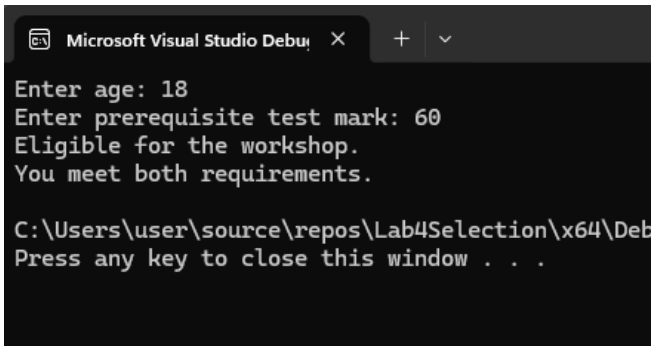
    if (!eligible)
    {
        cout << "Not eligible for the workshop." << endl;
```

```
    }
    return 0;
     }
```

(e) Test all four combinations below. Record the output and state which condition prevents eligibility in each unsuccessful case.

Age	Test mark	Expected output / reason
17	59	Not eligible for the workshop. Age and mark are both below the requirements.
17	60	Not eligible for the workshop. Age is below 18.
18	59	Not eligible for the workshop. Mark is below 60.
18	60	Eligible for the workshop. You meet both requirements.

OUTPUT FOR BOTH CONDITION MEET :



Exercise 6 | Optional: Monthly payment validator

Optional challenge: initialise a Boolean flag and guard division.

A shop lets customers divide a purchase into equal monthly payments, without interest. Write a C++ program that reads the **total purchase amount** and the **number of months** as a double and an integer, respectively.

(a) Declare a `bool` variable named `validPayment`. Assign it a logical expression that is true only when the total is **zero or greater** and months is **greater than zero**.

(b) When the input is valid, calculate `monthlyPayment = total / months` and display the payment to two decimal places.

(c) When the input is invalid, display `Invalid payment details.` Use two separate `if` statements. The division must occur only for valid input.

Total (RM)	Months	Expected result
1200.00	6	`Monthly payment = RM 200.00`
1200.00	0	`Invalid payment details.`
1200.00	-3	`Invalid payment details.`
-100.00	6	`Invalid payment details.`
0.00	6	`Monthly payment = RM 0.00`

(d) Explain why zero months must be rejected before the calculation.

Code for question 6 :

```
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    double total, monthlyPayment;
    int months;
    bool validPayment;

    cout << "Enter total purchase amount: RM ";
    cin >> total;

    cout << "Enter number of months: ";
    cin >> months;

    validPayment = (total >= 0) && (months > 0);

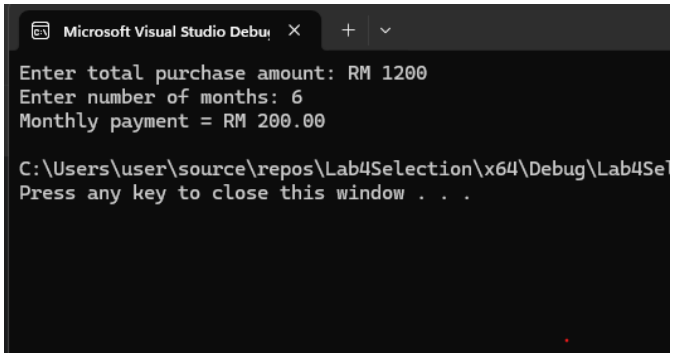
    if (validPayment)
    {
        monthlyPayment = total / months;

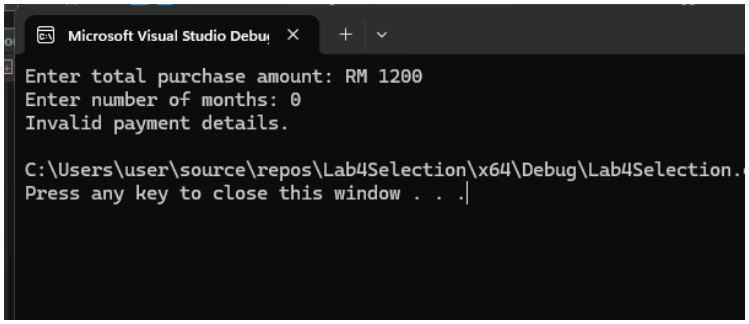
        cout << fixed << setprecision(2);
        cout << "Monthly payment = RM "
            << monthlyPayment << endl;
    }

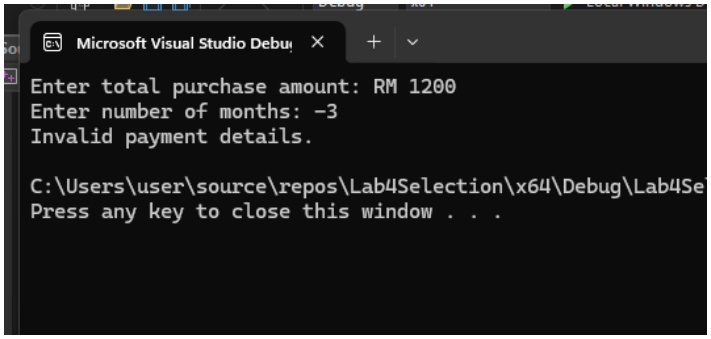
    if (!validPayment)
    {
        cout << "Invalid payment details." << endl;
    }

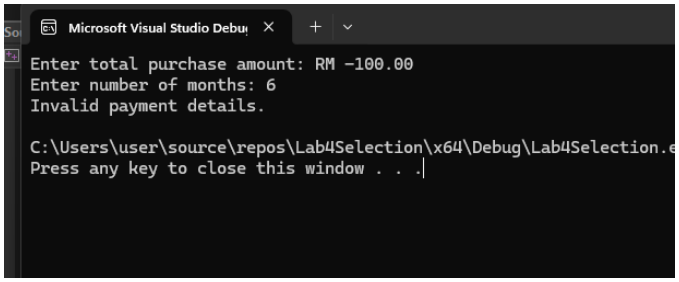
    return 0;
}
```

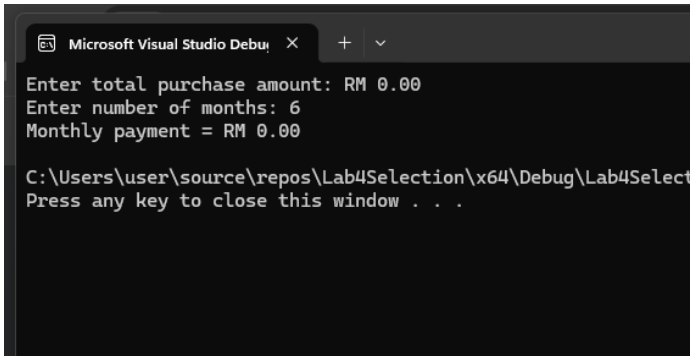
Test output :






Microsoft Visual Studio Debu
Enter total purchase amount: RM 1200
Enter number of months: -3
Invalid payment details.
C:\Users\user\source\repos\Lab4Selection\x64\Debug\Lab4Se
Press any key to close this window . . .


Microsoft Visual Studio Debu
Enter total purchase amount: RM -100.00
Enter number of months: 6
Invalid payment details.
C:\Users\user\source\repos\Lab4Selection\x64\Debug\Lab4Selection.e
Press any key to close this window . . .


Microsoft Visual Studio Debu
Enter total purchase amount: RM 0.00
Enter number of months: 6
Monthly payment = RM 0.00
C:\Users\user\source\repos\Lab4Selection\x64\Debug\Lab4Select
Press any key to close this window . . .